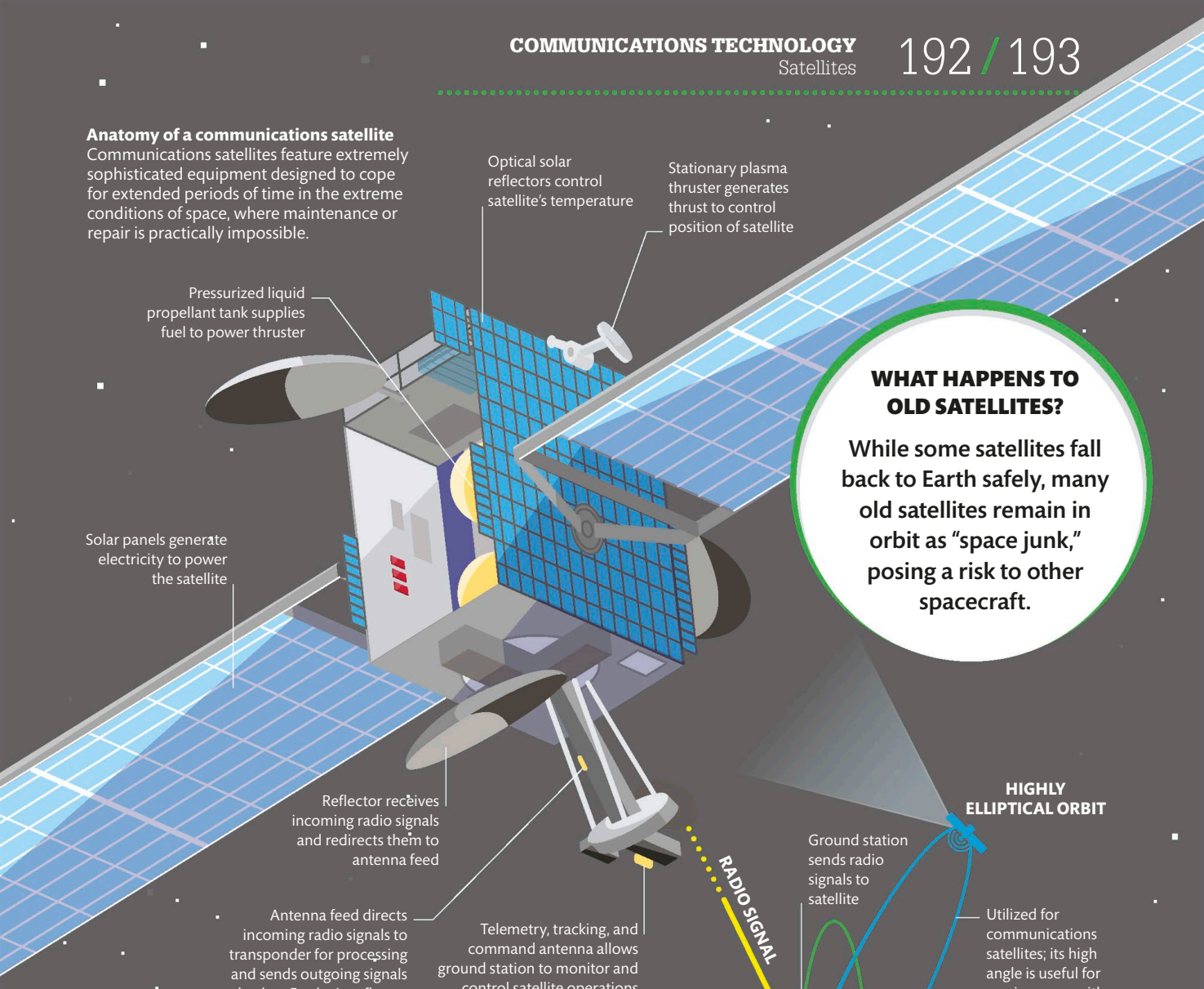
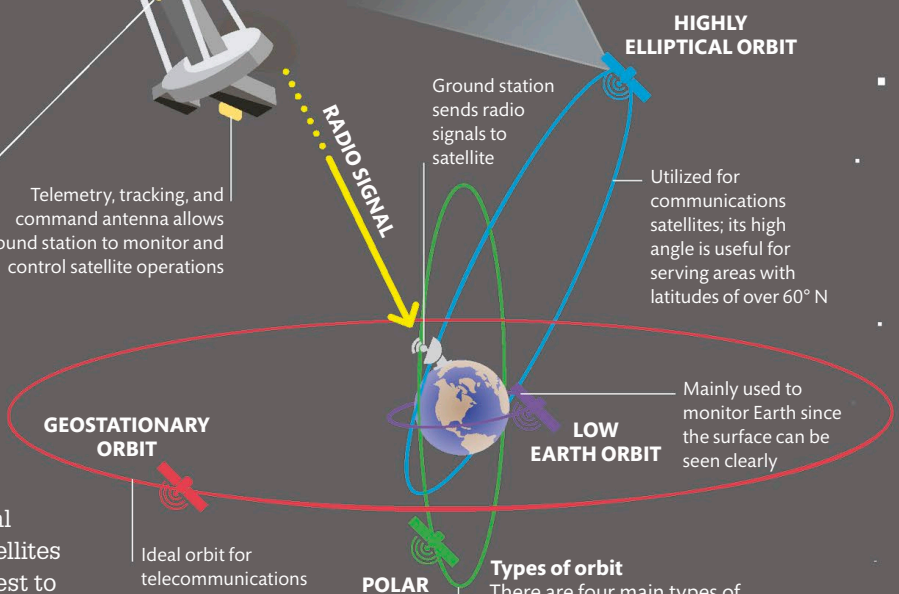
Anatomy of a communications satellite
Communications satellites feature extremely sophisticated equipment designed to cope for extended periods of time in the extreme conditions of space, where maintenance or repair is practically impossible.



WHAT HAPPENS TO OLD SATELLITES?

While some satellites fall back to Earth safely, many old satellites remain in orbit as "space junk," posing a risk to other spacecraft.



Satellite orbits

Satellites enter orbit if they launch at high enough velocity to overcome Earth's gravitational pull at the surface and then balance its weaker gravitational pull in space. Many communications satellites are in geostationary orbit. They travel west to east at the same rate as Earth spins, so they appear stationary above a point on the equator. Some satellites have polar orbits, crossing both poles on their journey around Earth.

Types of orbit
There are four main types of orbit around Earth, which are characterized by their shape, angle, and altitude. Most satellites are in low earth orbit, less than 1,250 miles (2,000 km) above the surface.